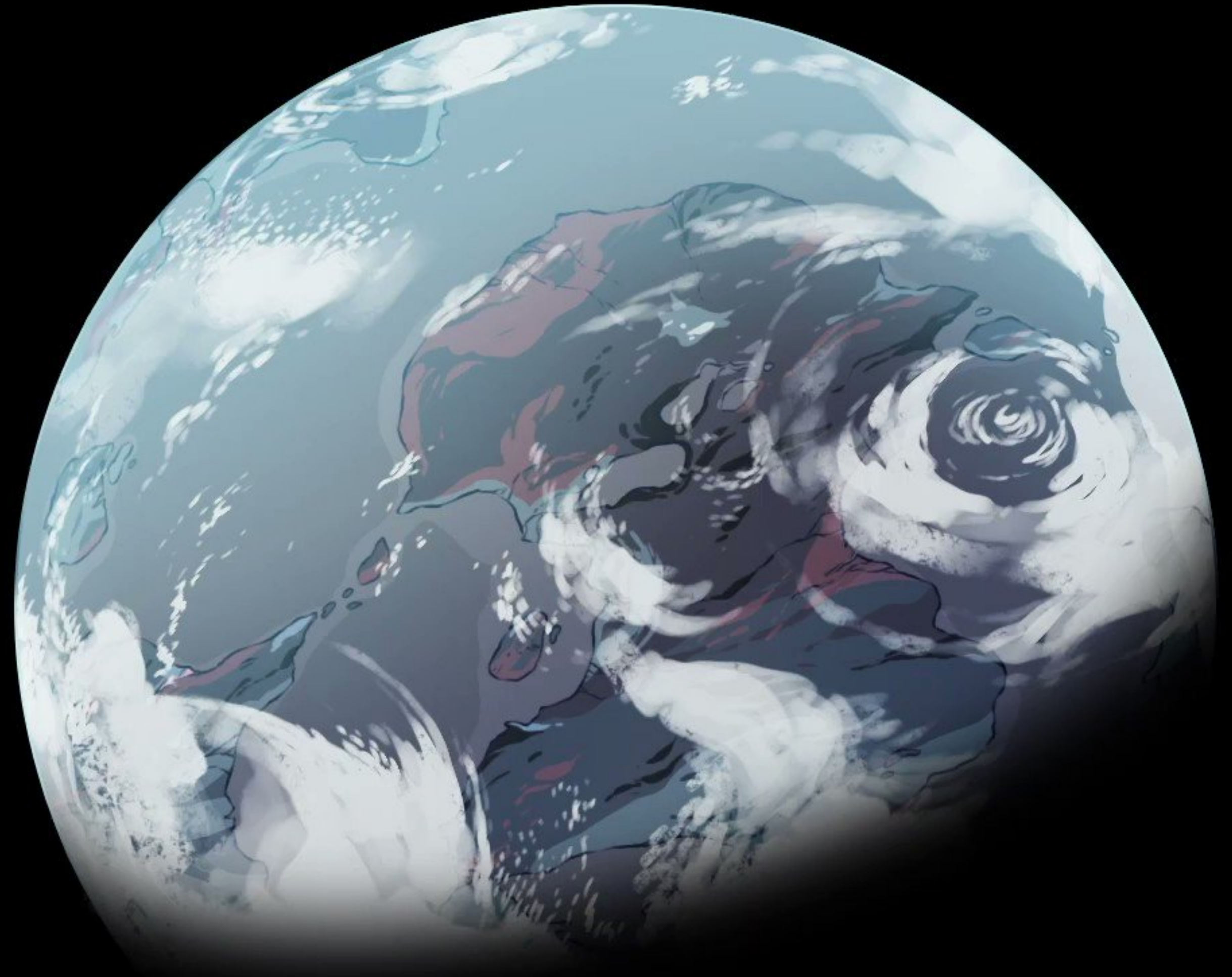


Science & Science Fiction

33-120 | Prof. Katie Breivik



Announcements

- Quiz 4 available **today**, due by **11:59pm**
- Exploration paper 4 due **March 28th at 11:59pm**
 - Topic: Extraterrestrial Intelligence; see Canvas for details
- Quiz 5 is coming up on **March 31st**
 - Topic: Extraterrestrial Intelligence from Chapter 5 & lecture material; 30 pts possible
- No class this Friday, March 28th

Conditions necessary for habitability

Planets/moons

“Habitable zone”
which allows for
liquid water

Moon that can
enable tidal pools

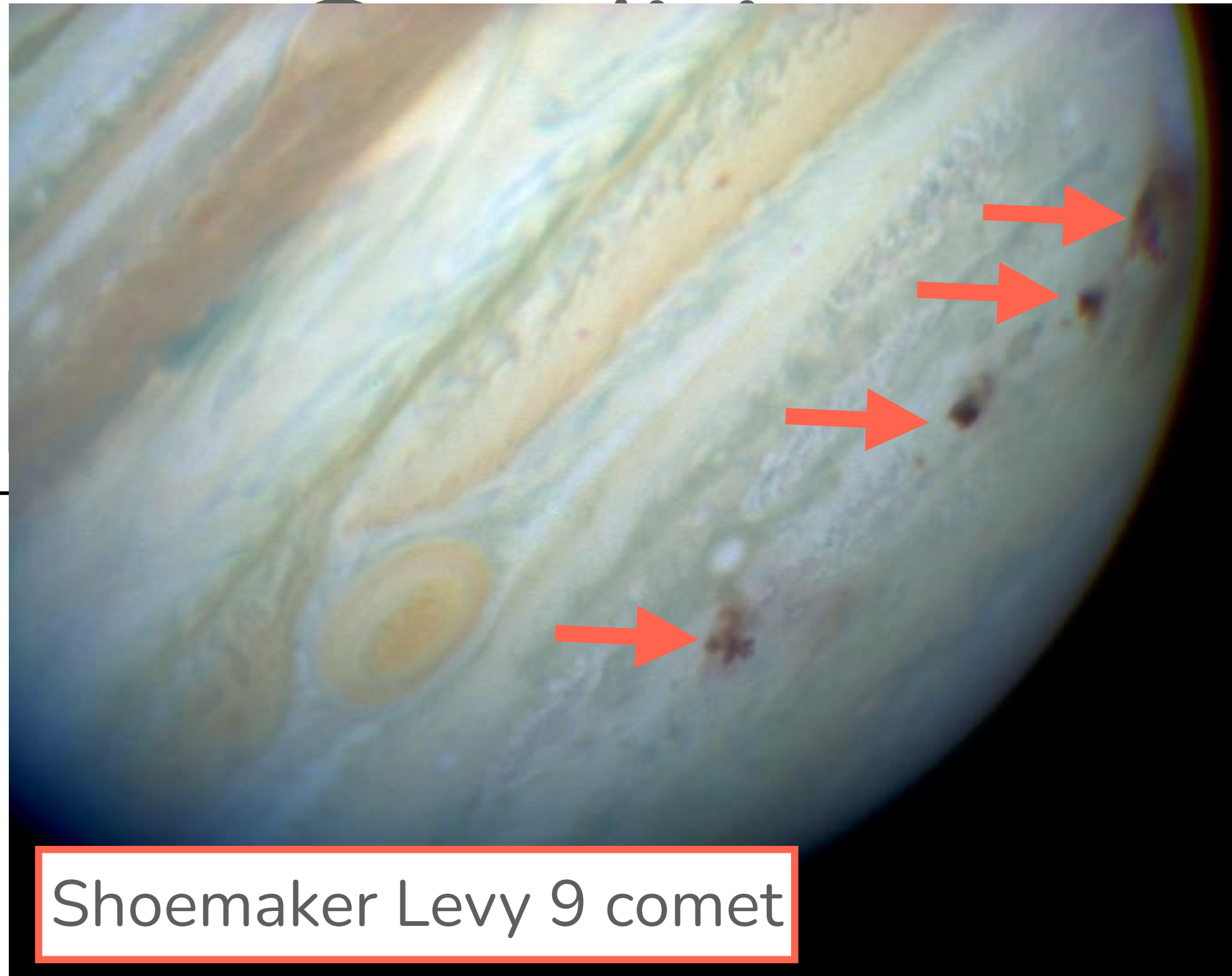
Stars

The Universe



Plumes of liquid water have been observed erupting out of Enceladus, a moon of Saturn that experiences strong tides that heat the moon

Necessary for habitability



Stars

Star must be old
enough for life to
evolve

The Universe

Moon that can
enable tidal pools

Gas giants which
protect inner rocky
planets from comets
& asteroids

Conditions necessary for habitability

Planets/moons

“Habitable zone”
which allows for
liquid water

Moon that can
enable tidal pools

Stars

Star must be old
enough for life to
evolve

Gas giants which
protect inner rocky
planets from comets
& asteroids

The Universe

Flat: perfect expansion rate
allows formation of stars &
galaxies

Matter imbalance: perfect
rate of 1 proton per 10^{10}
proton-antiproton pairs

Mass of proton is 1/1000
times **smaller** than mass of
neutron —> right β -decay rate
for stable elements

The Drake Equation

$$N = R^{\star} \times f_p \times n_E \times f_l \times f_i \times f_c \times L$$

$R^{\star}$: average star formation rate in the Galaxy (assumed 10 $\star$ /yr)

f_p : fraction of stars with planets (assumed < 1)

n_E : number of Earth-like planets per star (assumed > 1)

f_l : fraction of Earth-like planets with life (assumed ~ 1)

f_i : fraction of life sustaining planets with intelligent life (assumed ~ 1)

f_c : fraction of intelligent civilizations capable of communication (assumed ~ 1)

L : average lifetime of radio-communicating civilization (assumed..?)

The Drake Equation

$$N = R^{\star} \times f_p \times n_E \times f_l \times f_i \times f_c \times L$$

$R^{\star}$: average star formation rate in the Galaxy (**measured** $\sim 3 M_{\odot}/\text{yr} \rightarrow 6 \star/\text{yr}$)

f_p : fraction of stars with planets (**measured** ~ 1)

n_E : number of Earth-like planets per star (assumed > 1)

f_l : fraction of Earth-like planets with life (assumed ~ 1)

f_i : fraction of life sustaining planets with intelligent life (assumed ~ 1)

f_c : fraction of intelligent civilizations capable of communication (assumed ~ 1)

L : average lifetime of radio-communicating civilization (assumed..?)

The word "CONTACT" is displayed in a large, blue, three-dimensional serif font. The letters have a slight shadow and depth, giving them a 3D appearance. The text is centered within a solid black rectangular background.

Contact

Directed:
Robert Zemeckis

Warner Brothers (1997)

The Drake Equation

$$N = R^{\star} \times f_p \times n_E \times f_l \times f_i \times f_c \times L$$

$R^{\star}$: average star formation rate in the Galaxy (**measured** $\sim 3 M_{\odot}/\text{yr} \rightarrow 6 \star/\text{yr}$)

f_p : fraction of stars with planets (**measured** ~ 1)

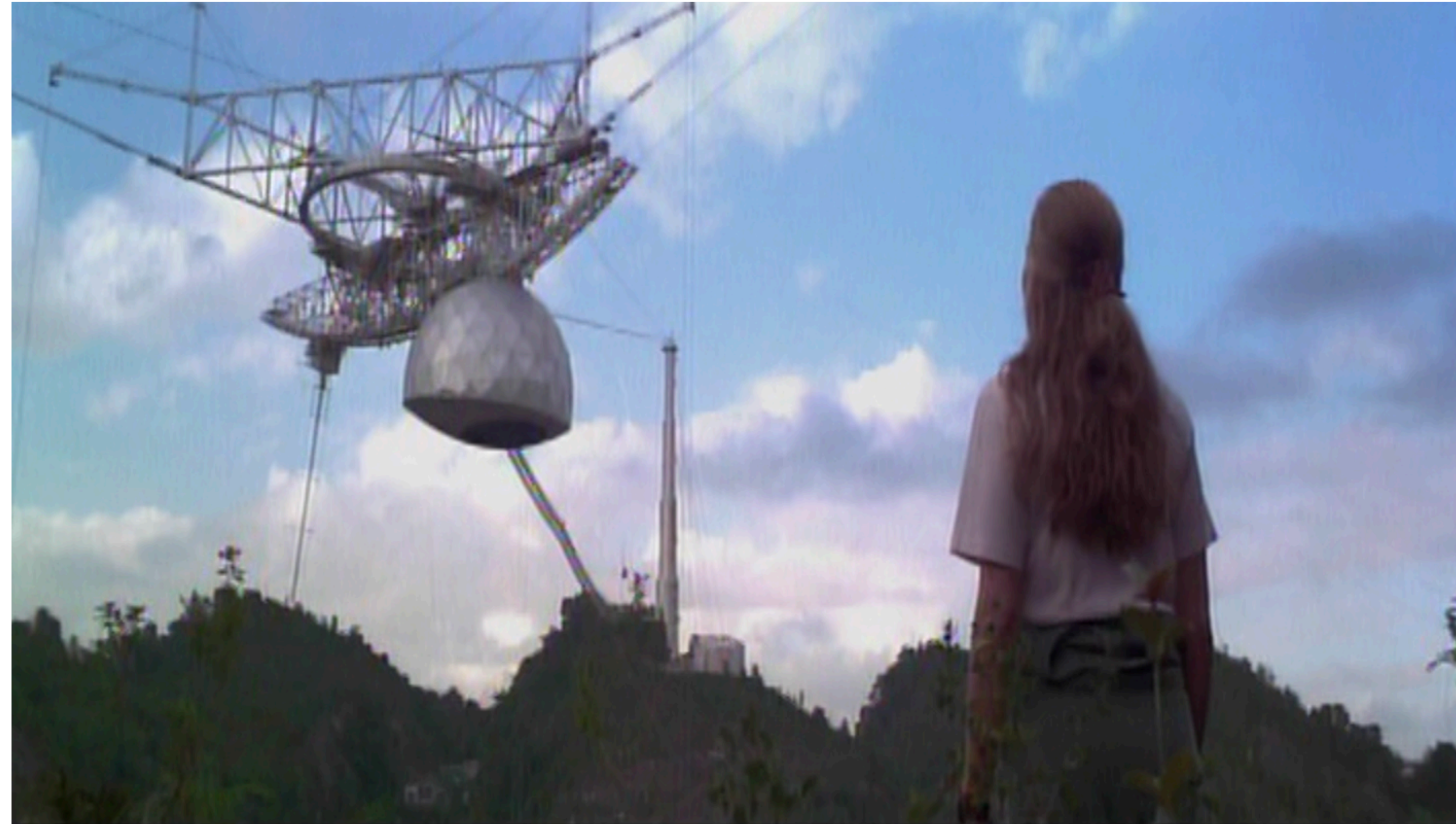
n_E : number of Earth-like planets per star (assumed > 1)

f_l : fraction of Earth-like planets with life (assumed ~ 1)

f_i : fraction of life sustaining planets with intelligent life (assumed ~ 1)

f_c : fraction of intelligent civilizations capable of communication (assumed ~ 1)

L : average lifetime of radio-communicating civilization (**?? Earth: ~ 100 years**)



Contact

Directed:
Robert Zemeckis

Warner Brothers (1997)

Arecibo observatory - Puerto Rico



305 m across &
completed in 1963

Largest radio dish
on Earth until FAST
dish was built by
China in 2016
(500 m)

Arecibo observatory - Puerto Rico



305 m across &
completed in 1963

Collapsed
December 1, 2020

Still out of
commission today



CONTACT

Contact

Directed:
Robert Zemeckis

Warner Brothers (1997)

There are 400 billion stars out there, just in our Galaxy alone

If only one out of a million of those had planets ...

And only one out of a million of those had life ...

And one out of a million of those had intelligent life ...

There would be *literally* millions of civilizations in the Galaxy

There are 400 billion stars out there, just in our Galaxy alone

If only one out of a million of those had planets ... $4 \times 10^{11} / 10^6 = 4 \times 10^5$

And only one out of a million of those had life ... $4 \times 10^5 / 10^6 = 0.4$

And one out of a million of those had intelligent life ... $0.4 / 10^6 = 4 \times 10^{-7}$

There would be *basically zero* of civilizations in the Galaxy

What could communication look like with ET?



Contact

Directed:
Robert Zemeckis

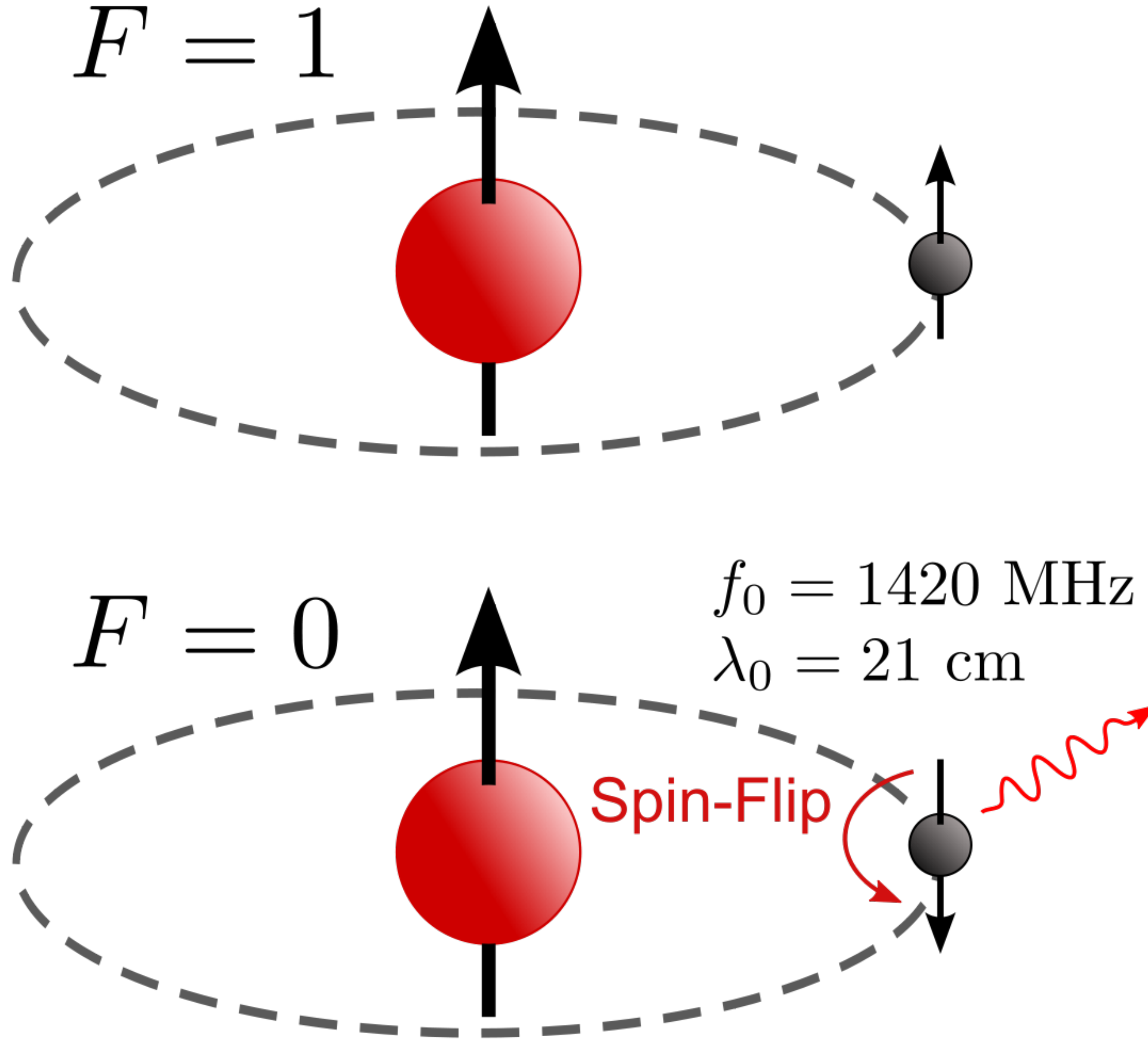
Warner Brothers (1997)

“Hydrogen times π ”

Hydrogen:
1 proton + 1 electron

Electrons have spin
energies that differ if
they are pointed
up or down

$$f_0 \times \pi = 1420 \text{ MHz} \times \pi = 4460 \text{ MHz}$$



Discussion question 26: <https://tinyurl.com/SciSciFi-26>



What could communication look like with ET?



Arrival

Directed:

Denis Villeneuve

Paramount (2016)

What could communication look like with ET?



Heptopods:

Human description of aliens
that visited Earth

Nonverbal

Communication:

Why did they choose to park
there?

Non-correlated communication between writing and sound

What could communication look like with ET?



Men In Black

Directed:
Barry Sonnenfeld

Columbia Pictures
(1997)